

Physiological Time-Series Signal Classification Using Continuous-Time Neural Networks

Final Report

Madhulika Vikraman: mvikrama@usc.edu

Shruti Rao: raoshrut@usc.edu

Vikash Churiwala: vchuriwa@usc.edu

Yi Yu: yyu421@usc.edu

Nakarin Korsuwan: korsuwan@usc.edu

Thanapat Trachu: trachu@usc.edu

Chancellor Roach : ccroach@usc.edu

Members Contributions

Member contributions were largely comparative across the project. Specific contributions are detailed by members below.

Yi : ECG Project: Literature Review, Data pre-processing, Baseline, continuous time model building and model performance comparison, Drafting slides for the ECG part.

Madhulika: ECG Project: Literature Review, Liquid Neural Network and all other Continuous-time model implementation (without pre-processing, keeping temporal patterns), drafting slides for concept and ECG results

Roach: ECG Project: Literature review, Liquid Neural Network research, put together products (Proposal, Midterm, and Final reports).

Thanapat: EEG Project: Liquid Neural Network and all other Continuous-time model implementation for EEG, Implementation of Baselines for EEG including ResNet18, LSTM, and CNN.

Nakarin: EEG Project: Literature review, Data pre-processing, Data Splitting, Resampling Sequential TS, Preparing Pytorch Dataset for Training, Preparing slides for EEG part.

Shruti: ECG Project: Literature Review, Data Splitting at sample level, baseline models Feedforward NN, LSTM, LTC, model evaluation and comparison, Identification and analysis of data leakage in evaluation methodology

Vikash: EEG Project: Literature review, LNN baseline model implementation along with pipeline, data preprocessing for handling overfitting in LNN

Abstract

This paper investigates the applicability of continuous-time neural network architectures to physiological time-series signal classification, focusing on EEG-based seizure prediction and ECG-based myocardial infarction detection. Because EEG and ECG signals are continuous, noisy, and non-stationary, continuous-time models such as Liquid Time-Constant Networks, Continuous-Time Recurrent Neural Networks, Neural Ordinary Differential Equations, and Continuous-Time Gated Recurrent Units may offer advantages over traditional discrete-time deep learning approaches. This report compares these models against conventional baselines including CNN-LSTM, CNN-Transformer, ResNet, CNN, and LSTM architectures. Results show that continuous-time models achieved competitive performance on ECG classification and strong in-domain EEG performance, while requiring substantially fewer parameters than several traditional baselines. However, out-of-domain EEG results declined significantly, indicating that patient-level generalization remains a major challenge. Overall, the findings suggest that continuous-time models are promising for efficient biomedical signal classification, but further work is needed to improve robustness, imbalance handling, and clinical generalizability.

Introduction

Electrical signals generated by the human body as seen in EEGs and ECGs possess complex and crucial health information and are non-stationary, noisy, complex and continuous, reflecting numerous biological processes influenced both by hidden internal states and apparent external conditions of the human body. Continuous-Time based deep learning models are more suited to prediction in multi-variate time-series signal data as they adapt to systems with adaptive dynamics. Their ability to evolve internal states, continuously determined by ordinary differential equations (ODEs) or time-aware decay mechanisms, makes them particularly useful for modeling time series data in a variety of applications. This project attempts to explore how useful Continuous-Time based models can serve as universal approximators when compared to traditional methods.

This paper aims to systematically study Continuous-Time approaches like Liquid Time Constant Networks (LTCs), Continuous-Time Recurrent Neural Networks (CTRNNs), Neural Ordinary Differential Equation models (NODEs) and Continuous-Time Gated Recurrent Units (CTGRUs) for medical electrical signal processing, investigating the expressibility of such networks against multiple time series prediction tasks specifically in predicting seizures and heart conditions evolving over time and comparing them against current state-of-the-art models designed for the same tasks. The following sections cover a brief literature review that encompasses standard techniques in the field and flows into how models were employed and finalized with a results and discussion section.

Literature Review

The varying ways to process and understand varying types of data has been a topic of interest in the research world for decades. As new methods for machine learning, deep learning, and other data processing techniques are developed at ever increasing rates, it becomes difficult to understand how each new method applies to various different types of tasks. Electrical signals generated by the human body as seen in EEGs and ECGs are one such unique example of data used in these tasks. This study analyzes the current standard methods for analyzing and predicting information from EEGs and ECGs and then looks at the rarely used Continuous-Time models and how they could be used in seizure and heart-disease prediction .

Continuous-Time Models

Continuous-time approach: In traditional sequential Machine or Deep Learning methods like RNNs and LSTMs, time flows in discrete steps like $t = 1, 2, 3, 4$. The updates to the layers also happen in fixed steps according to $h_t = f(h_{t-1}, x_t)$. In a continuous model, updates happen at every instant according to $\frac{dh(t)}{dt} = f(h(t), x(t))$. EEG and ECG signals are naturally continuous, with irregular samples and different patterns, hence we try to use Continuous-time models to match the nature of data, and capture natural patterns.

Liquid Neural Networks : Liquid Neural Networks are a major example of Continuous-time models. They are neural networks whose behavior

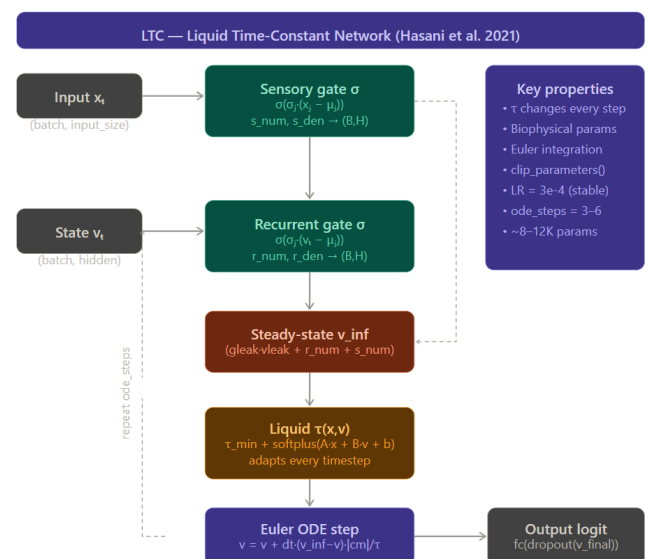
continuously adapts over time, like a liquid changing shape depending on its environment. Where a regular Neural Network follows a rigid pipeline (input $\rightarrow$ fixed layers $\rightarrow$ output) and their internal behavior is mostly static after training, LNNs behave differently, using continuous-time dynamics where each neuron has a state that changes continuously over time, controlled by differential equations, and the network can adjust its own speed and response. The network is adaptive and flexible (“liquid”) and can handle irregular time-series naturally.

Liquid Time Constant Networks : While LNN is the general idea of a Neural Network with time-varying dynamics, LTC is the specific architecture that implements the concept. Liquid Time-Constant Networks provide one of the clearest theoretical motivations for modeling complex biomedical signals with continuous-time dynamics. An LTC is a continuous-time neural network where each neuron has a dynamic, learnable time constant (τ) that controls the speed of reaction to inputs. Where all neurons update at the same fixed rate in RNNs, some neurons have fast reflexes or slow ones in LTCs. A small τ reacts quickly to capture spikes, and larger τ reacts slowly to capture trends. Thus, they are designed to adapt to fast and slow patterns automatically.

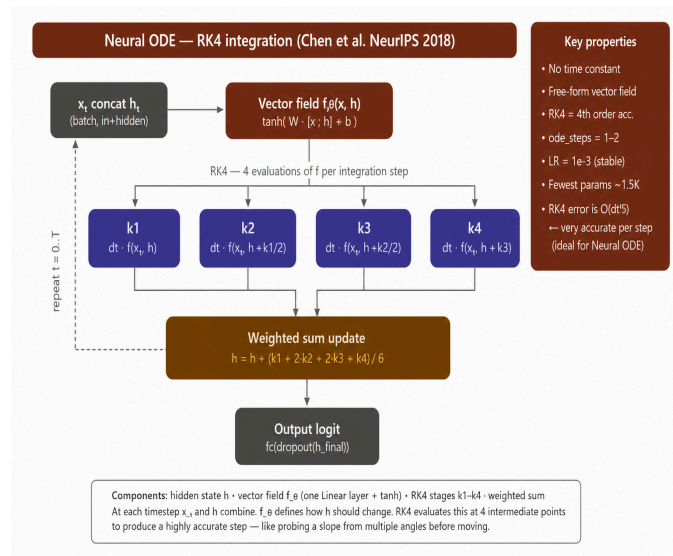
Hasani et al. (2021) describe LTCs as recurrent neural models whose hidden states evolve according to input-dependent ordinary differential equations, allowing the system’s time constants to vary dynamically with the input. They argue that this formulation improves expressivity while maintaining bounded and stable dynamics. In their time-series experiments, LTCs outperform several standard recurrent and continuous-time baselines on many tasks, suggesting that adaptive continuous-time state evolution may be advantageous when temporal dynamics are complex. At the same time, the paper is careful to note limitations, including vanishing-gradient issues for long-term dependencies, sensitivity to the numerical solver, and higher time and memory costs than simpler continuous-time alternatives. These tradeoffs are important because they suggest LTCs are promising, but not automatically superior in every sequence-learning setting (Hasani et al., 2021).

A broader continuous-time perspective is provided by Fitzgerald et al. (2023), who review and evaluate CTRNN-style architectures for irregularly sampled medical time series. Their central claim is that many medical sequences are not naturally observed at regular intervals, making continuous-time hidden-state evolution conceptually attractive. In their blood-glucose forecasting experiments, ODE-LSTM and related architectures often outperform standard discrete-time GRU or LSTM baselines on several metrics, indicating that continuous evolution between observations can be useful. However, their findings are cautious: the models remain sensitive to measurement error, require sufficient data, and are often outperformed by gradient-boosted tree methods (GBTs) (Fitzgerald et al., 2023).

Taken together, these two continuous-time papers above, provide a strong conceptual rationale for investigating LTCs in physiological signal analysis. EEG and ECG data are both temporally evolving, noisy, and often non-stationary, making them plausible candidates for adaptive continuous-time modeling. Both papers strongly emphasize the need to compare any results with other baseline candidates for the applicable use case. An additional approach this report explores the internal comparison between different Continuous-Time models like Liquid Time Constant Networks, Neural ODEs (Ordinary Differential Equations), CTRNNs (Continuous Time Recurrent Neural Networks) and CTGRUs (Continuous Time Gated Recurrent Units), and their collective performance against traditional Deep Learning Techniques.

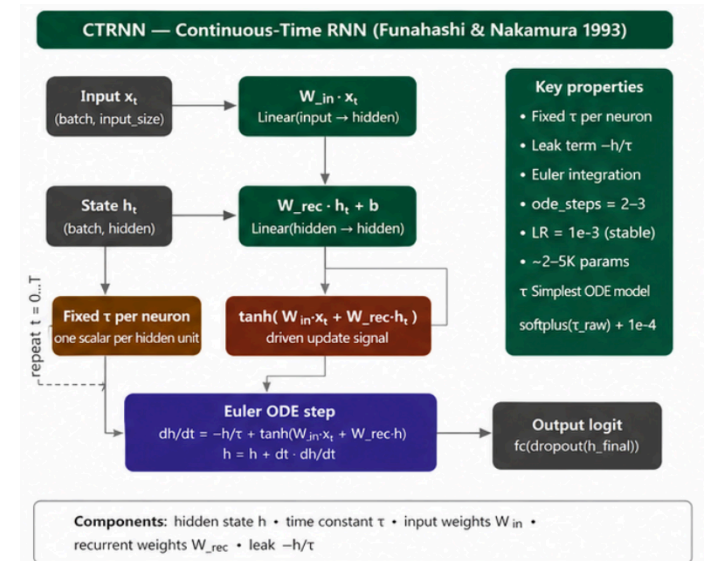


Neural Ordinary Differential Equations : a new family of deep neural network models. Instead of specifying a discrete sequence of hidden layers, NODEs parameterize the derivative of the hidden state using a neural network $\frac{\delta h(t)}{\delta t} = f(h(t), t, \theta)$ and computes the output by passing this through a black-box ODE solver. NODE models the entire network as a continuous transformation. In this formulation, the neural network parameters (θ) determine how the state changes at each point in time. These continuous depth models have constant memory cost via the adjoint sensitivity method, and the ability to adapt its computation to each input's complexity. Neural ODEs can be understood as continuous-time control systems, where their ability to interpolate data can be interpreted in terms of controllability, and are used in time series analysis tasks. It has no time constant, the function $f(h, t)$ implicitly controls the speed. For our task, Neural ODEs are theoretically attractive because ECG/EEG signals are the output of a continuous dynamical system (the brain and heart's electrophysiology). While training Chen et. Al (2018) shows how to scalably backpropagate through any ODE solver, without access to its internal operations. This allows end-to-end training of ODEs within larger models.



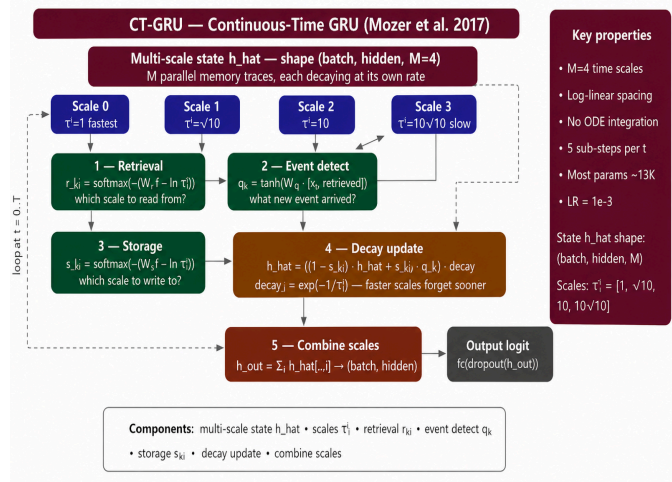
Continuous Time Recurrent Neural Networks : a type of artificial neural network where the hidden state paths are governed by ordinary differential equations (ODEs) and operate in continuous time,

rather than the discrete time-step updates used in traditional RNNs. The CTRNN dynamics are described by: $\frac{\delta u_i}{\delta t} = \frac{-u_i}{\tau_i} + \sum_1^j w_{ij} \sigma(u_j) + I_i(t)$, where each neuron i has a time constant τ_i governing how quickly it forgets its state and how fast it responds (Funahashi et al. 1993) proved that CTRNNs are universal approximators of dynamical systems, that any finite-time trajectory of any smooth dynamical system can be approximated by a CTRNN's internal states. CTRNNs use fixed time constants i.e. τ is fixed and not learned dynamically. Each neuron has a fixed speed of response and its behaviour is static after training. Thus, CTRNNs are continuous, but rigid. For our task, dynamic biomedical modeling, CTRNNs seem to be well suited due to their continuous nature and input dependent internal states.



Continuous Time Gated Recurrent Units : A CTGRU is a GRU adapted for continuous time by incorporating time decay into its memory updates. A regular GRU updates memory at each step, but CTGRU considers how much time has passed and updates memory accordingly, i.e. forgetting based on time gaps. Mozer et al. (2017) extends the GRU to handle event sequences and replaces the GRU storage and retrieval gates with storage and retrieval scales, computed from the external input and the current hidden state. The scale is expressed in terms of a time constant. The update gate learns to distribute a new event across these traces via a softmax weighting, and crucially the traces decay

exponentially between events as $e^{\frac{-\Delta t}{\tau_i}}$, making time a first-class citizen rather than just another input feature. It adds exponential decay of hidden state and time aware updates. Large Δt (time gap) means more forgetting and small Δt means less forgetting and can thus handle irregular timesteps in the data. For EEG/ECG continuous time tasks, it is hypothesized that the CT-GRU may perform better than a vanilla GRU due to its ability to capture the multivariate features in the time-series analysis tasks, despite both of them having similar performance metrics in the paper.



EEG

In seizure-prediction literature, EEG modeling is consistently framed as difficult because of noise, non-stationarity, and especially inter-patient variability. Alarfaj et al. (2025) emphasize that seizure patterns differ substantially from one patient to another, which makes generalized models difficult to optimize. To address this, they propose an Ensemble of Deep Transfer Learning (EDTL) framework that combines pretrained ResNet and EfficientNet backbones with a custom 2DCNN for personalized seizure detection. Their method first converts EEG signals into spectrogram-like representations using short-time Fourier transforms and sliding windows, then fine-tunes the model for patient-specific detection. Their results show strong performance, with the ensemble achieving an AUC of 99.23% across the evaluated datasets. Importantly, their work suggests that personalization and transfer learning may be more valuable than simply scaling a single generic model, especially in the presence of high inter-patient variability (Alarfaj et al., 2025).

Afsari et al. (2025) offer a complementary perspective by systematically comparing machine-learning and deep-learning methods for seizure prediction across different feature representations and prediction windows. Their results show that engineered-feature approaches remain highly competitive, particularly at longer pre-ictal windows: a DeepConvNet with engineered features achieved 94.10% accuracy and 88.03% sensitivity at a 30-minute prediction window. By contrast, a CNN-LSTM hybrid trained on raw EEG performed best at the shorter 5-minute window, reaching 94.98% accuracy. This is a significant finding because it shows that the superiority of handcrafted features versus raw-signal learning depends on the temporal horizon of the prediction problem. In other words, this resource conveys that there is not one straightforward best method to implement and changes based on the unique system design and the wanted prediction window (Afsari et al., 2025).

Together, the EEG papers suggest that seizure modeling is moving toward deeper, more adaptive systems, but not toward a complete abandonment of domain knowledge. Alarfaj et al. (2025) still rely on STFT-based transformation and patient-specific adaptation, while Afsari et al. (2025) show that handcrafted features remain powerful in some prediction settings. This implies that successful EEG models often combine data-driven learning with structured signal representations rather than relying purely on raw end-to-end architectures.

ECG

From the ECG literature, it can be seen that Atwa et al. (2025) proposed interpretable deep learning models for arrhythmia classification that combine waveform-based convolutional processing with demographic features such as age and gender. Their study highlights several persistent challenges in ECG modeling, including class imbalance, noise and artifacts, high dimensionality, generalizability across clinical settings, and the black-box nature of deep networks. Their results indicate that contextual demographic information can improve classification performance, especially for underrepresented classes, while lead-level interpretability analysis can provide clinically useful insight into which ECG channels are most influential in the model's decisions. A key feature is that it frames performance and interpretability not as competing

goals, but as parallel requirements for clinically credible ECG data analysis systems (Atwa et al., 2025).

A deeper analysis into Atwa et al. (2025) shows that the dominant trend in ECG analysis is toward hybrid architectures. They summarize prior work using CNN-LSTM combinations, transformer-enhanced methods, entropy-informed signal representations, transfer learning, and attention-based beat fusion. Across these studies, the common logic is that ECG signals contain both biological patterns and longer temporal dependencies, and therefore benefit from models that combine convolutional feature extraction with sequential or attention-based mechanisms. This hybridization trend closely parallels what is seen in seizure modeling and reinforces the idea that physiological signals are rarely well-served by overly simplistic single-mechanism models (Atwa et al., 2025).

Across both EEG and ECG studies that were looked at, several common themes appear. First, hybridization has become the norm: CNN-RNN, CNN-LSTM, attention-based models, transfer-learning ensembles, and multi-branch architectures dominate recent work because they capture complementary temporal and spatial information (Alarfaj et al., 2025; Afsari et al., 2025; Atwa et al., 2025). Second, representation still matters. Although deep learning reduces dependence on manual feature design, spectrograms, STFT-derived inputs, entropy measures, and other structured representations they still continue to play an important role, particularly when signal characteristics are subtle or class imbalance is severe (Alarfaj et al., 2025; Afsari et al., 2025; Atwa et al., 2025). Third, personalization and transfer learning are increasingly important, especially in cases where patient-specific variation is a major obstacle to generalization (Alarfaj et al., 2025).

Literature Conclusion

There exists a meaningful gap in literature concerning the applications of LTC networks. While there exists numerous models on more standard deep learning and machine learning methods such as CNN's, LSTMs, GRUs etc, LTCs lack a significant presence in similar fields. This is increasingly meaningful because of their theoretical applicability when it comes to interpreting

continuous time based physiological signals. LTCs have the capability to provide tangible advantages over strong hybrid baselines on biomedical electrical signals. This project will attempt to empirically explore the feasibility of this application of LTCs.

Methodology

The scope and goals of this project have not changed significantly from the proposal. This project systematically studies Liquid Neural Networks and other Continuous-Time Networks for medical electrical signal processing, investigating their predictive abilities against multivariate time series prediction tasks, specifically in predicting seizures and heart conditions over time and comparing them against traditional and current state-of-the-art models designed for the same tasks. The work completed can be split into three sections covering work with LNN models as well as work done in building benchmarks to compare against both concerning EEG and ECG fields. Each of these sections below details the data sets being used, methods employed and is finalized with a synopsis of results and a discussion.

Continuous-time Employed Methods

We will be seeing the performance of Liquid Time Constant Networks (LTCs), Continuous-Time Recurrent Neural Networks (CTRNNs), Neural Ordinary Differential Equation models (NODEs) and Continuous-Time Gated Recurrent Units (CTGRUs) and comparing their performance with each other as well as against the traditional machine learning and deep learning methods.

Data Sets Being Used

PTB Diagnostic ECG Database : The PTB Diagnostic ECG Database is a high-resolution ECG dataset hosted by PhysioNet. It contains 549 ECG recordings from 290 subjects, including healthy controls and patients with various cardiac diseases. Each record includes 15 simultaneously recorded ECG signals: the conventional 12 leads and three Frank leads. The signals were sampled at 1000 Hz with 16-bit resolution. Most records also include clinical summaries containing demographic information, diagnosis, medical history, medications, interventions, and additional cardiac examination data.

PTB-XL ECG Database: The PTB-XL dataset is a large publicly available 12-lead ECG dataset containing 21,837 ten-second ECG recordings from 18,885 patients. Each record includes standard 12-lead ECG waveform data, available at both 100 Hz and 500 Hz sampling rates, along with clinical metadata and cardiologist-provided annotations. The annotations follow the SCP-ECG standard and include diagnostic, rhythm, and morphology statements, making the dataset suitable for ECG classification, arrhythmia detection, abnormality screening, and deep learning-based cardiovascular diagnosis.

Model Approach

The LNN section of this project can be approached by viewing four different variations of the LNN concept. Below details a short synopsis of the status of current working models that have been developed and details can be found in the appendix.

- **LTC** - Tested an LTC on both pre-processed and unprocessed PTB-XL dataset as training and validation dataset and the PTB Diagnostic dataset as test dataset.
- **CTRNN** - Tested a CTRNN on both pre-processed and unprocessed PTB-XL dataset as training and validation dataset and the PTB Diagnostic dataset as test dataset. The ODE is discretised with Euler steps, number of steps = 3.
- **CTGRU** - Tested a simple CTGRU on both pre-processed and unprocessed PTB-XL dataset as training and validation dataset and the PTB Diagnostic dataset as test dataset. It maintains 4 separate memory traces each with a different decaying timescale.
- **NODE** - Tested a simple NODE model on both pre-processed and unprocessed PTB-XL dataset as training and validation dataset and the PTB Diagnostic dataset as test dataset. It implements a small MLP integrated with Euler's steps, number of steps = 3.

EEG Employed Methods

This section describes the methodology employed for EEG-based seizure prediction. It includes a

description of the dataset, preprocessing procedure, feature preparation, and the model used for classification.

Data Set Being Used

Training Datasets for Epilepsy Analysis: Preprocessing and Feature Extraction from EEG Time Series:

The Zenodo dataset *Training Datasets for Epilepsy Analysis: Preprocessing and Feature Extraction from EEG Time Series* contains 20 patient-specific CSV training datasets derived from EEG recordings of epileptic patients from the Freiburg Seizure Prediction Database. The dataset provides preprocessed EEG-derived feature matrices rather than raw EEG waveforms. For each patient, 1,080 univariate and bivariate features were extracted using a sliding-window method with two-second windows and one- or two-second time shifts. Each window includes a binary label indicating seizure or non-seizure activity, making the dataset suitable for supervised machine learning tasks such as epileptic seizure detection and prediction.

Data Preprocessing

The dataset contains EEG signals from 20 epileptic patients. Each feature has already been extracted using a 2-second window of 256 Hz resulting in 512 Hz per timestep. The dataset also has metadata related to how they are extracted such as registration and actual TAG. However, this report is only interested in the actual TAG since it contains the ground truth to predict.

The main goal for the EEG dataset is to do seizure prediction which aims at identifying the seizure before the IKTAL state happens. Therefore, this study is only interested in 3 types of actual TAG which are PRE and PRE_IKTAL (positive class) and INTER (negative class). Due to the high volume of the dataset, 3 patients are selected randomly; 1 patient as out-of-domain test set and the other two for splitting into 3 sets; training set, validation set and test set with 80%, 10% and 10% proportion respectively, resulting in 7417, 927, 928 and 3453 samples in training, development, test and out-of-domain test set, respectively. Stratified group splitting has been applied based on registration which is the identification of the number from the Freiburg EEG database and only on seen patients. It attempts to make the number of positive classes to be all split which in turn will lower the issue of

imbalance dataset. The next step is data cleaning. Besides the simple clean up, the main focus is to figure out the common feature that all the patients have so that constant features can be eliminated (i.e., feature with variance equal to 0). This process is necessary because before modelling all the features need to be normalized and with zero variance some features can be NaNs. After normalization, the data needs to be sequenced so that they could be used in the sequential modelling such as LSTMs or RNNs. The sequence length was chosen to be 300. The value is about 10 minutes and is related to the typical advance time for predicting seizure. Also, the stride is chosen to be 1 in order to utilize all the dataset completely.

Sequencing is also based on each registration. As a result, each sample will have 300 sequence length and 1008 features and will be ready for batching when the model is being trained next.

Patient	# Patients	Split	# Samples	# PRE class	# INTER class
Seen	3	Training	7684	1194	6490
		Validation	991	221	770
		Test	698	38	660
		Total	9373	1453	7920
Unseen	1	Test	3862	1243	2619

Model Approach

There are three baselines for this task. Two of them are adopted from Alarfaj et al. (2025). The first baseline is a ResNet18 model pretrained on ImageNet, a large-scale image classification dataset with around 1000 classes. The architecture consists of multiple convolutional layers with residual connections to prevent gradient vanishing or exploding. To adapt the model to our task, the first convolutional layer is replaced to accept single-channel input instead of three channels. The final classification layers are also replaced with a linear layer that outputs a single value followed by a sigmoid activation. To reduce overfitting and preserve the pretrained representation, all original layers are frozen except the modified ones.

Although pretrained models are effective for transfer learning, they are often large and computationally expensive. For example, ResNet18 has around 11 million parameters, which makes deployment on low-resource devices difficult. To address this, a simpler convolutional neural network is introduced. This model contains five convolutional layers, each followed by a ReLU

activation and a max-pooling layer. The feature dimension increases progressively, starting from 32 and doubling at each layer up to 512. An average pooling layer is applied before flattening, and the resulting features are passed through two linear layers that reduce the dimension to a single output. This model is trained from scratch.

In addition to convolution-based models, a different architecture is explored using a long short-term memory (LSTM) network. The model consists of two LSTM layers with a hidden dimension of 128, followed by a classification head with two linear layers that reduce the representation to a single output. Before classification, average pooling over the time dimension is applied. This model is also trained from scratch using the same training procedure.

All models are trained for 10 epochs using binary cross-entropy loss. The Adam optimizer is utilized with a learning rate of 0.001. Early stopping is applied with a patience of 20 epochs to prevent overfitting.

ECG Employed Methods

Data Set Being Used

[PTB Diagnostic ECG Database](#): Description covered above under LNN methods.

Dataset	Split	Total beats	Number of MI beats	Total participants	Number of MI subjects
PTB_XL	Train	11539	4382	10438	3744
	Validation	2893	1104	2610	936
	Total	14432	5486	13048	4680
PTB_Diagnosis	Test	4862	3924	200	148

Data Preprocessing

This study follows the preprocessing steps described by Kachuee et al., (Kachuee et al., 2018) as follows: 1) Down-sampling the original 1 kHz ECG signal to 125 Hz, 2) Splitting ECG signal to 10s windows, 3) Normalizing the data between 0-1, 4) Finding all the R-peaks from the 10s window 5) Select the median of R-R interval as the representative heartbeat with the period T 6)Extent the two R peaks to 1.2T 6)Padding the signal to 150 samples (corresponding to 4s).

Machine Learning Task

A primary focus was to test the binary classification task (Myocardial infarction vs healthy) performance of the models, which is one of the major tasks in the field. However, multi-class classification was employed as well to differentiate between different types of the myocardial diseases may be included. After preprocessing, 4862 heartbeats samples from 200 subjects were acquired(148 with MI, 52 are controls).

Two different dataset splitting methods were tested. One is based on the subject, and the other is based on the heartbeats. For the first data splitting method, heartbeats samples were randomly selected from 160 subjects in the training dataset, and the remaining in the test datasets. For the second data splitting method, 3889/4862 heartbeats samples were selected in the training dataset, and the remaining 973/4862 heart beats in the test dataset.

Model Approach

For the benchmark model: The focus is to evaluate two neural network architectures for binary ECG classification including a CNN-LSTM and a CNN Transformer model. The CNN-LSTM architecture combines convolutional layers for automated feature extraction from raw ecg beats with LSTM layers to capture temporal dependencies and sequential patterns in the extracted features. The CNN-Transformer model replaces the LSTM component with a transformer encoder which identifies important patterns across the signal without relying on sequential processing. Both baseline models are trained and evaluated using subject level data splitting to prevent data leakage, which is important to consider since all heart beats from a given patient subject are assigned to either the training or test set. To address class imbalance in the dataset consisting of 80.9% patient cases and 19.1% healthy controls, both models use focal loss which applies higher weights to minority class examples and emphasizes cases that are harder to classify during training.

Results

ECG Results

Baseline

Below is a table displaying the baseline analysis results undergoing the tests and modeling described above. The CNN-LSTM model has the highest ECG test F1 of 0.84.

	CNN-LSTM		CNN- Transformer	
	Val (PTBXL)	Test (PTB_Diagnosis)	Val	Test
Sensitivity	0.72	0.81	0.72	0.80
Specificity	0.94	0.94	0.91	0.90
Accuracy	0.86	0.85	0.84	0.83
F1	0.84	0.83	0.83	0.81

Continuous-Time Results With Preprocessing

	LTC		CTRNN		NODE		CT-GRU	
	Val	Test	Val	Test	Val	Test	Val	Test
Sensitivity	0.73	0.80	0.74	0.79	0.71	0.79	0.75	0.77
Specificity	0.93	0.88	0.92	0.88	0.95	0.90	0.93	0.88
Accuracy	0.86	0.82	0.86	0.82	0.86	0.82	0.86	0.80
F1	0.84	0.79	0.84	0.79	0.84	0.79	0.85	0.77

The continuous-time models were slightly lower compared to benchmarks, but remained competitive - LTC, CTRNN, and NODE each achieved 0.82 test accuracy and 0.79 F1, while CT-GRU achieved 0.80 accuracy and 0.77 F1. These results suggest that continuous-time architectures can approach the performance of stronger conventional baselines on ECG classification, although they didn't outperform the best CNN-LSTM model in this setting.

Continuous Time Results Without Preprocessing

	LTC		CTRNN		NODE		CT-GRU	
	Val	Test	Val	Test	Val	Test	Val	Test
Sensitivity	0.83	0.77	0.89	0.95	0.84	0.95	0.90	0.95
Specificity	0.81	0.85	0.85	0.92	0.88	0.90	0.89	0.95
Accuracy	0.82	0.85	0.87	0.92	0.86	0.90	0.90	0.95
F1	0.78	0.33	0.84	0.55	0.83	0.49	0.87	0.66

When continuous-time models were trained without the preprocessing pipeline, performance improved substantially in terms of accuracy for several models. CT-GRU achieved the strongest test accuracy at 0.95, followed by CTRNN at 0.92 and NODE at 0.90. This suggests that continuous-time models may benefit from preserving longer temporal structure rather than reducing the ECG into preprocessed beat-level segments. However, the F1 scores remained considerably lower than the accuracy values, indicating that class imbalance, decision-threshold effects, or metric averaging may be affecting model evaluation. Therefore, these results should be interpreted as promising but requiring further validation.

EEG Results

The result on the EEG dataset has been divided into in-domain settings (on seen patients) and out-of-domain settings (on unseen patients).

In-domain setting

	Accuracy	Precision	Recall	F1
ResNet18	94.99%	0.685	1.000	0.521
CNN	99.86%	0.987	1.000	0.974
LSTM	100.00%	1.000	1.000	1.000
LTC	99.57%	0.962	1.000	0.927
CTRNN	100.00%	1.000	1.000	1.000
CTGRU	100.00%	1.000	1.000	1.000
NODE	99.43%	0.947	0.947	0.947

In this experiment, the model is evaluated on samples from patients included in the training set. The results suggest that all models can learn seizure patterns for these patients. According to the result, the performance of both the baselines and continuous models are high, achieving the accuracy of almost 100%. This suggests that the models successfully captured the seizure pattern of each patient in the training set. However, due to the limited resources we could do the experiment on more patients in order to ensure that the performance is not likely to be from an overfitting.

Out-of-domain setting

	Accuracy	Precision	Recall	F1
ResNet18	41.25%	0.501	0.918	0.345
CNN	61.52%	0.485	0.564	0.426
LSTM	54.40%	0.505	0.723	0.388
LTC	66.39%	0.416	0.372	0.472
CTRNN	34.90%	0.474	0.910	0.320
CTGRU	33.45%	0.478	0.946	0.320
NODE	67.06%	0.122	0.071	0.429

In this setting, samples are used from a new patient who is not included in the training set as the test set. This setup evaluates the model's ability to generalize to an unseen patient. Compared to the in-domain setting, the performance of all models degrades significantly. According to the table, the performance of the baselines and some of the continuous models, including LTC and NODE are similar. This suggests that they have the same level of generalization. However, the model size of the continuous models are extremely smaller compared to the baselines model. For example, the largest continuous model, which is CTGRU, has 138K parameters, while the smallest baselines, which is LSTM, has 700K parameters. This makes the continuous models more efficient in terms of the number of parameters than the non-continuous baselines, allowing a deployment on even a low-resource device.

Low precision in all the models could suggest the problem in the imbalance dataset. Mostly, all the models see only the normal period where seizure does not occur and try to make a guess on the prediction. In order to address the problem, we could add training data with more positive classes or apply an alternative loss function such as focal loss.

Discussion

Overall, the results suggest that continuous-time neural networks are a promising but not automatically superior approach for biomedical electrical signal classification. For ECG myocardial

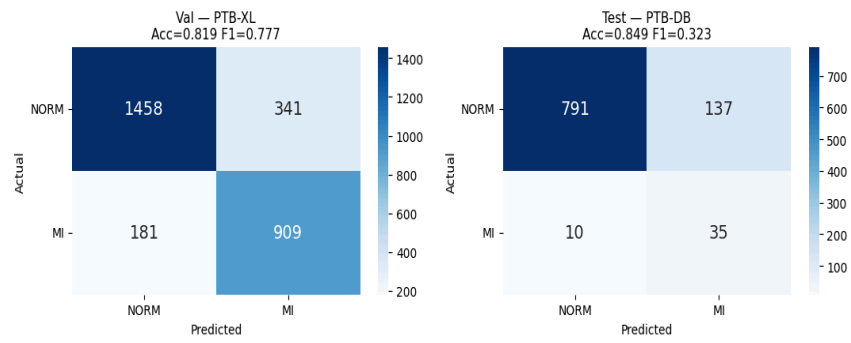
infarction detection, the continuous-time models achieved performance close to the CNN-LSTM and CNN-Transformer baselines, and the non-preprocessed ECG experiments suggest that these models may benefit from preserving the full temporal structure of the signal rather than relying only on heartbeat-level preprocessing. However, the difference between high accuracy and lower F1 scores in some ECG results indicates that class imbalance, thresholding, and metric choice must be considered carefully. For EEG seizure prediction, all models performed extremely well in the in-domain setting, suggesting that both traditional and continuous-time architectures can learn seizure-related patterns from patients represented in the training data. However, performance dropped substantially in the out-of-domain setting, showing that generalization to unseen patients remains a

major challenge. This suggests that the models may be learning patient-specific patterns rather than fully generalizable seizure signatures. Continuous-time models did not consistently outperform traditional baselines across all metrics, but they achieved competitive results with far fewer parameters, making them attractive for low-resource biomedical applications such as wearable or embedded monitoring systems. Overall, these experiments support the feasibility of continuous-time models for ECG and EEG signal classification, but future work should focus on larger patient-level validation, improved imbalance handling, increasing the number of ODE steps, threshold calibration, and evaluation methods that better reflect real clinical deployment conditions.

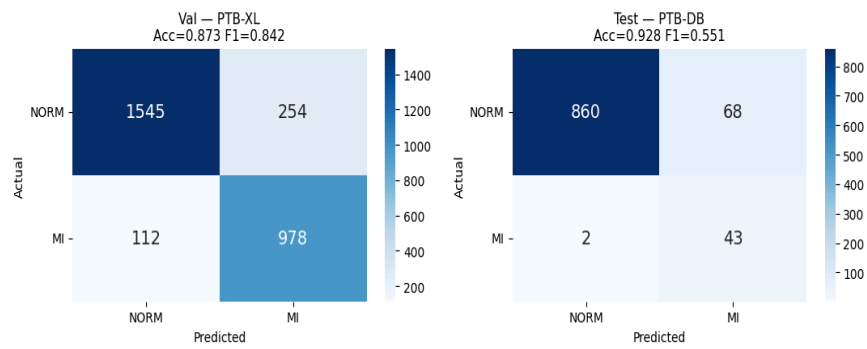
Appendix

ECG Confusion Matrices

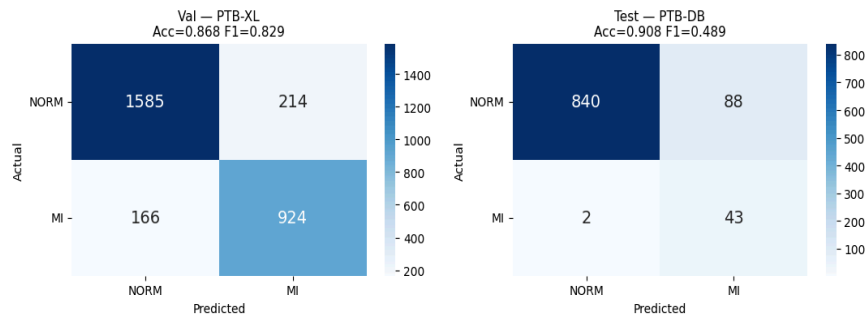
LTC



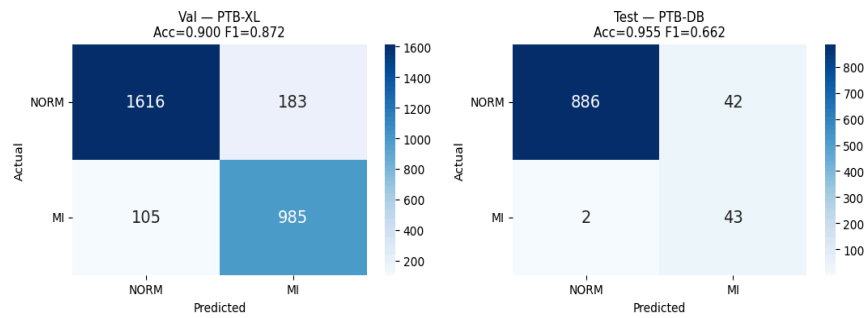
CTRNN



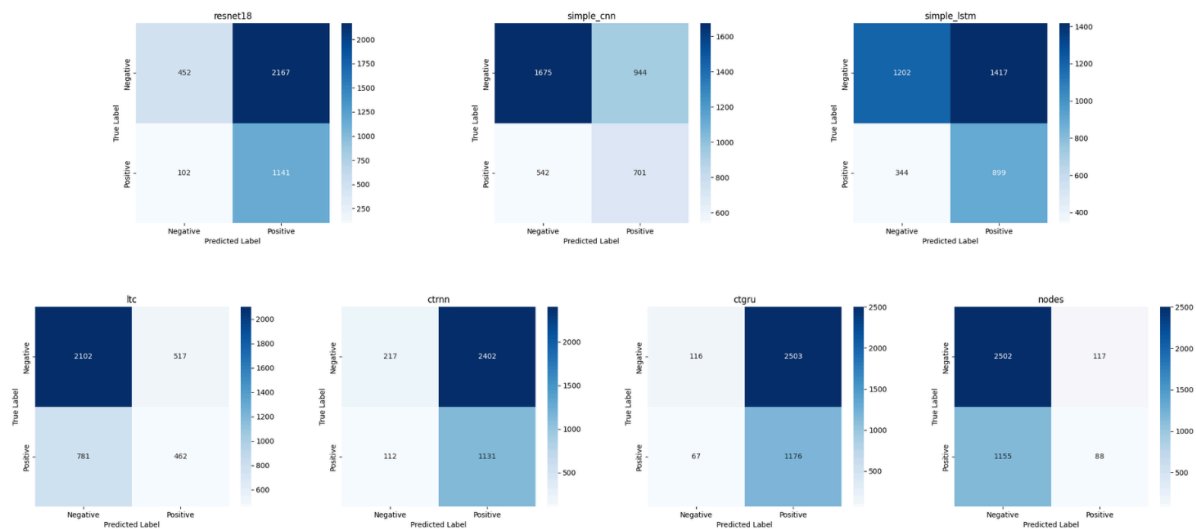
NODE



CT-GRU



EEG Confusion Matrices



Generative AI Usage

Generative AI has been used in this project in the following ways:

- Used as a code assist for writing and debugging models, and analyze results and potential gaps in the approaches
- Utilized to summarize some portions of information for the literature review (and then heavily edited)
- To build the works cited references in MLA format (works themselves were **not** found through AI)
- Was used in writing the abstract and discussion sections (after providing take-aways)

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